

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

PRACTICAL-1:

Aim: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait ()`.
5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Procedure :

```
venkatsaikrishna5@VSK-HP: ~  
venkatsaikrishna5@VSK-HP:~$ mkdir OSSP  
mkdir: OSSP: File exists  
venkatsaikrishna5@VSK-HP:~$ cd OSSP  
venkatsaikrishna5@VSK-HP:~/OSSP$ nano command_exec.c  
venkatsaikrishna5@VSK-HP:~/OSSP$ gcc command_exec.c -o command_exec  
venkatsaikrishna5@VSK-HP:~/OSSP$ ls  
command_exec.c.save  command_exec.c  
command_exec         command_exec.c.save  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: uname  
  
Parent Process  
Parent PID: 3669  
  
Child Process  
Child PID: 3670  
Parent PID: 3669  
Linux  
Child process completed.  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: lscpu  
  
Parent Process  
Parent PID: 3680  
  
Child Process  
Child PID: 3681  
Parent PID: 3680  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:         39 bits physical, 48 bits virtual  
Byte Order:            Little Endian  
CPU(s):                12  
On-line CPU(s) list:   0-11  
Vendor ID:             GenuineIntel  
Model name:            13th Gen Intel(R) Core(TM) i5-1335  
U  
CPU family:            6  
Model:                 186  
Thread(s) per core:    2
```

```
venkatsaikrishna5@VSK-HP: ~$ cat /proc/cpuinfo

Model:                               186
Thread(s) per core:                  2
Core(s) per socket:                  6
Socket(s):                           1
Stepping:                            3
BogoMIPS:                            4991.99
Flags:                                fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat ps
e36 clflush mmx fxsr sse sse2 ss h
t syscall nx pdpe1gb rdtscp lm con
stant_tsc rep_good nopl xtopology
tsc_reliable nonstop_tsc cpuid tsc
_known_freq pni pclmulqdq vmx ssse
3 fma cx16 pcid sse4_1 sse4_2 x2ap
ic movbe popcnt tsc_deadline_timer
aes xsave avx f16c rdrand hypervi
sor lahf_lm abm 3dnowprefetch ssbd
ibrs ibpb stibp ibrs_enhanced tpr
_shadow ept vpid ept_ad fsgsbase t
sc_adjust bmi1 avx2 smep bmi2 erms
invpcid rdseed adx smap clflushop
t clwb sha_ni xsaveopt xsavec xget
bv1 xsaves avx_vnni vnm1 umip wait
pkg gfni vaes vpclmulqdq rdpid mov
diri movdir64b fsrm md_clear seria
lize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:                      VT-x
Hypervisor vendor:                   Microsoft
Virtualization type:                 full
Caches (sum of all):
L1d:                                 288 KiB (6 instances)
L1i:                                 192 KiB (6 instances)
L2:                                  7.5 MiB (6 instances)
L3:                                  12 MiB (1 instance)
NUMA:
NUMA node(s):                        1
NUMA node0 CPU(s):                   0-11
Vulnerabilities:
```

```
venkatsaikrishna5@VSK-HP: ~$ cat /proc/cpuinfo

Vulnerabilities:
Gather data sampling:                Not affected
Ghostwrite:                          Not affected
Indirect target selection:           Not affected
Itlb multihit:                      Not affected
L1tf:                                Not affected
Mds:                                 Not affected
Meltdown:                            Not affected
Mmio stale data:                     Not affected
Old microcode:                       Not affected
Reg file data sampling:               Mitigation; Clear Register File
Retbleed:                            Mitigation; Enhanced IBRS
Spec rstack overflow:                Not affected
Spec store bypass:                   Mitigation; Speculative Store Bypa
ss disabled via prctl
Spectre v1:                          Mitigation; usercopy/swapgs barrie
rs and __user pointer sanitization
Spectre v2:                          Mitigation; Enhanced / Automatic I
BRS; IBPB conditional; PBRSE-eIBRS
SW sequence; BHI BHI_DIS_S
Srbds:                               Not affected
Tsa:                                 Not affected
Tsx async abort:                     Not affected
Vmscape:                             Not affected

Child process completed.
venkatsaikrishna5@VSK-HP: ~/OSSP$ ./command_exec
Enter a Linux command: lsblk

Parent Process
Parent PID: 3693

Child Process
Child PID: 3694
Parent PID: 3693
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
```

```
venkatsaikrishna5@VSK-HP: ~  
Child process completed.  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: ps  
  
Parent Process  
Child Process  
Parent PID: 3704  
Child PID: 3705  
Parent PID: 3704  
PID TTY TIME CMD  
3578 pts/0 00:00:00 bash  
3704 pts/0 00:00:00 command_exec  
3705 pts/0 00:00:00 ps  
Child process completed.  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: top  
  
Parent Process  
Parent PID: 3715  
  
Child Process  
Child PID: 3716  
Parent PID: 3715  
top - 06:04:56 up 52 min, 1 user, load average: 0.00, 0.00, 0.00  
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7796.6 total, 7001.0 free, 643.8 used, 311.8 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7152.8 avail Mem  


| PID | USER     | PR | NI | VIRT  | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|-------|-------|-------|---|------|------|---------|-----------------|
| 1   | root     | 20 | 0  | 24144 | 14448 | 10604 | S | 0.0  | 0.2  | 0:01.75 | systemd         |
| 2   | root     | 20 | 0  | 3180  | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.03 | init-systemd(Ub |
| 6   | root     | 20 | 0  | 3180  | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 49  | root     | 19 | -1 | 50400 | 16812 | 15524 | S | 0.0  | 0.2  | 0:00.39 | systemd-journal |
| 84  | systemd+ | 20 | 0  | 22540 | 14628 | 12012 | S | 0.0  | 0.2  | 0:00.14 | systemd-resolve |
| 93  | root     | 20 | 0  | 35360 | 12352 | 9268  | S | 0.0  | 0.2  | 0:02.23 | systemd-udev    |
| 113 | root     | 20 | 0  | 2888  | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.06 | chronyd-starter |
| 114 | root     | 20 | 0  | 4472  | 3104  | 2860  | S | 0.0  | 0.0  | 0:00.02 | cron            |
| 116 | message+ | 20 | 0  | 8792  | 5492  | 4732  | S | 0.0  | 0.1  | 0:00.18 | dbus-daemon     |


```

```
venkatsaikrishna5@VSK-HP: ~  
top - 06:05:05 up 52 min, 1 user, load average: 0.00, 0.00, 0.00  
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7796.6 total, 7012.0 free, 632.9 used, 311.8 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7163.8 avail Mem  


| PID  | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|------|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 3577 | root     | 20 | 0  | 3200   | 1256  | 1108  | S | 0.3  | 0.0  | 0:00.05 | Relay(3578)     |
| 1    | root     | 20 | 0  | 24144  | 14448 | 10604 | S | 0.0  | 0.2  | 0:01.75 | systemd         |
| 2    | root     | 20 | 0  | 3180   | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.03 | init-systemd(Ub |
| 6    | root     | 20 | 0  | 3180   | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 49   | root     | 19 | -1 | 50400  | 16812 | 15524 | S | 0.0  | 0.2  | 0:00.39 | systemd-journal |
| 84   | systemd+ | 20 | 0  | 22540  | 14628 | 12012 | S | 0.0  | 0.2  | 0:00.14 | systemd-resolve |
| 93   | root     | 20 | 0  | 35360  | 12352 | 9268  | S | 0.0  | 0.2  | 0:02.23 | systemd-udev    |
| 113  | root     | 20 | 0  | 2888   | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.06 | chronyd-starter |
| 114  | root     | 20 | 0  | 4472   | 3104  | 2860  | S | 0.0  | 0.0  | 0:00.02 | cron            |
| 116  | message+ | 20 | 0  | 8792   | 5492  | 4732  | S | 0.0  | 0.1  | 0:00.18 | dbus-daemon     |
| 159  | root     | 20 | 0  | 44092  | 29780 | 14648 | S | 0.0  | 0.4  | 0:00.40 | networkd-dispat |
| 163  | root     | 20 | 0  | 18492  | 9528  | 8224  | S | 0.0  | 0.1  | 0:00.21 | systemd-logind  |
| 177  | syslog   | 20 | 0  | 220548 | 5892  | 4624  | S | 0.0  | 0.1  | 0:00.12 | rsyslogd        |
| 216  | root     | 20 | 0  | 5492   | 2712  | 2528  | S | 0.0  | 0.0  | 0:00.01 | agetty          |
| 217  | _chrony  | 20 | 0  | 20420  | 6384  | 5608  | S | 0.0  | 0.1  | 0:00.14 | chronyd         |
| 232  | _chrony  | 20 | 0  | 12092  | 2448  | 1804  | S | 0.0  | 0.0  | 0:00.00 | chronyd         |
| 246  | root     | 20 | 0  | 123456 | 32620 | 18004 | S | 0.0  | 0.4  | 0:00.34 | unattended-upgr |
| 318  | root     | 20 | 0  | 8648   | 5516  | 4672  | S | 0.0  | 0.1  | 0:00.02 | login           |
| 384  | venkats+ | 20 | 0  | 21936  | 11380 | 9176  | S | 0.0  | 0.1  | 0:00.37 | systemd         |
| 386  | venkats+ | 20 | 0  | 22912  | 4112  | 2112  | S | 0.0  | 0.1  | 0:00.00 | (sd-pam)        |
| 412  | venkats+ | 20 | 0  | 6136   | 5416  | 3864  | S | 0.0  | 0.1  | 0:00.04 | bash            |
| 828  | polkitd  | 20 | 0  | 306444 | 8444  | 7628  | S | 0.0  | 0.1  | 0:00.21 | polkitd         |
| 3576 | root     | 20 | 0  | 3184   | 1116  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 3578 | venkats+ | 20 | 0  | 6268   | 5544  | 3872  | S | 0.0  | 0.1  | 0:00.11 | bash            |
| 3715 | venkats+ | 20 | 0  | 2768   | 1820  | 1720  | S | 0.0  | 0.0  | 0:00.00 | command_exec    |
| 3716 | venkats+ | 20 | 0  | 8172   | 6052  | 3948  | R | 0.0  | 0.1  | 0:00.03 | top             |


```

Code:

```
venkatsaikrishna5@VSK-HP: - x + v
GNU nano 8.7.1 command_exec.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Invalid command!\n");
        exit(1);
    }
    else {
        // Parent Process
        printf("\nParent Process\n");

        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo      M-6 Copy

```

```
venkatsaikrishna5@VSK-HP: - x + v
GNU nano 8.7.1 command_exec.c
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Invalid command!\n");
        exit(1);
    }
    else {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo      M-6 Copy

```

Observation :

- The parent and child processes were successfully created using `fork()`, and their PIDs were displayed.
- The child successfully executed the user-entered Linux command using `exec()`.
- The parent process waited for the child using `wait()`.
- `uname`, `lscpu`, and `lsblk` displayed operating system, CPU, and storage information.
- `ps` and `top` displayed running processes and resource usage.
- The experiment showed that the operating system manages hardware resources and provides services that allow applications to use CPU, memory, storage, and I/O devices without directly accessing the hardware.

Result :

The experiment successfully demonstrated **Linux command execution using `fork()`, `exec()`, and `wait()`** and investigated the relationship between hardware resources and OS services. The Linux operating system was observed to provide effective abstractions and management of **CPU, memory, storage, processes, and I/O resources**.